VENANCE KOMU

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EDUCATION

Alfred University — Alfred, NY **Expected May 2029**

B.S. Electrical Engineering & Mathematics (Double Major), Minor in Computer Science | GPA: 3.95/4.00

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Organization & Architecture, Probability & Statistics, Linear Algebra, Investments & Managerial Finance, MATLAB

Honors & Awards: Dean's List (2025 & 2026) | Top 5, MAT Math Competition (Tanzania) | SAT Math: 790/800

PROFESSIONAL EXPERIENCE

Alfred University — Alfred, NY **May 2026 – July 2026**

Undergraduate Researcher, Secondary Sources of Gallium

- Developed a stochastic Material Flow Analysis (MFA) model mapping gallium flows across U.S. imports, manufacturing, consumption, and recycling, using **Monte Carlo** simulation to quantify supply uncertainty, identify untapped recovery opportunities, and assess pathways to reduce import dependence and strengthen supply-chain resilience.
- Engineered Python (pandas, NumPy) pipelines to clean, merge, and statistically compare secondary gallium-source datasets with XRF concentration measurements, assessing candidate feedstocks' viability for gallium recovery.

National Housing Corporation — Moshi, Tanzania **May 2025 – July 2025**

Intern, Financial Analysis & Software Development

- Analyzed **$500K+** in company funds, building Excel-based financial forecasting and cash-flow models of rental revenue and maintenance costs, and identified cost-optimization opportunities that contributed to a **4%** reduction in operating expenses.
- Developed Python data-processing pipelines to automate financial data cleaning, validation, and reporting, leveraging SQL queries to reconcile transactions, detect payment discrepancies, and improve financial reporting accuracy and consistency.

TECHNICAL PROJECTS

Market Liquidity & Investment Risk Intelligence Engine **June 2026 – Present**

C++17, Python, STL, CMake, Linux, Git

- Built a **C++17** event-driven market simulation engine supporting order matching, partial fills, and cancellations, reconstructing order-book activity and tracking market depth and order-flow imbalance to estimate execution costs.
- Implemented a Python risk-analysis pipeline translating bid-ask spread, liquidity, and execution-cost metrics into investor-facing alerts; validated engine behavior through automated tests covering matching priority and order-book updates.

PCB Component Detection & Critical-Material Recovery Engine **June 2026 – Present**

Python, PyTorch, YOLOv8, NumPy, Monte Carlo Simulation

- Trained a YOLOv8 object-detection model in PyTorch across 8 PCB component classes, achieving **85.97% mAP@50**, and integrated its predictions into an automated board-level critical-material recovery analysis pipeline for e-waste.
- Developed a probabilistic framework using 10,000 Monte Carlo simulations and variance-based sensitivity analysis to quantify recovery uncertainty across 5 critical materials; validated recovery calculations with 6 passing unit tests.

TCMS Utility Billing Portal **September 2023 – November 2023**

PHP, Laravel, REST APIs

- Developed a Laravel (PHP) billing portal managing tariffs, customer accounts, meters, debts, and payments via a REST API.
- Enforced per-action role-based access control (Spatie) and built role-specific dashboards showing customer, debt, and token totals.

LEADERSHIP & TECHNICAL DEVELOPMENT

ColorStack, Alfred University Chapter — Alfred, NY | *Treasurer* **August 2026 – Present**

- Manage a **$20K** club budget, analyzing expenditures and building Excel cost forecasts that identified **$4K** in projected savings on workshops and career events, reporting regularly to the executive board.

freeCodeCamp | *Machine Learning & Scientific Computing with Python* **July 2026 – October 2026**

- Built TensorFlow/Keras models for CNN image classification, SMS spam detection, and healthcare cost regression, plus a scikit-learn KNN book recommender on **1.1M** ratings and a Rock Paper Scissors algorithm.

SKILLS & INTERESTS

Languages: C++ (C++17, STL), Python, C, SQL, Java, JavaScript, MATLAB, VHDL

Tools & Frameworks: Linux, Git/GitHub, CMake, NumPy, pandas, Matplotlib, PyTorch, scikit-learn, TensorFlow/Keras, OpenCV, Excel

Interests: Chess, Puzzles, Ping Pong, Gymnastics, Pickleball